Journal of Dentistry 155 (2025) 105601

Contents lists available at ScienceDirect

Journal of Dentistry

journal homepage: www.elsevier.com/locate/jdent

Prediction of pink esthetic score using deep learning: A proof of concept

Ziang Wu [a,b,c,d,e,1], Yizhou Chen [f,1], Xinbo Yu [b,c,d,g], Feng Wang [h], Haochen Shi [i], Fang Qu [a,b,c,d,e], Yingyi Shen [a,b,c,d,e], Xiaojun Chen [i,**], Chun Xu [a,b,c,d,e,*]

[a] Department of Prosthodontics, Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, 200011, China
[b] College of Stomatology, Shanghai Jiao Tong University, Shanghai, China
[c] National Center for Stomatology and National Clinical Research Center for Oral Diseases, Shanghai, China
[d] Shanghai Key Laboratory of Stomatology and Shanghai Research Institute of Stomatology, Shanghai, China
[e] Shanghai Engineering Research Center of Advanced Dental Technology and Materials, Shanghai, China
[f] Department of Nuclear Medicine, University of Bern, Bern, Switzerland
[g] Second Dental Center, Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China
[h] Division of Oral and Maxillofacial Surgery, Faculty of Dentistry, The University of Hong Kong, 34 Hospital Road, Sai Ying Pun, Hong Kong SAR, China
[i] Institute of Biomedical Manufacturing and Life Quality Engineering, State Key Laboratory of Mechanical System and Vibration, School of Mechanical Engineering, Shanghai Jiao Tong University, Shanghai, 200240, China

ARTICLE INFO

Keywords:
Artificial intelligence
Dental esthetics
Deep learning
Dental implants
Implant dentistry

ABSTRACT

Objectives: This study aimed to develop a deep learning (DL) model for the predictive esthetic evaluation of single-implant treatments in the esthetic zone.

Methods: A total of 226 samples, each comprising three intraoral photographs and 12 clinical features, were collected for proof of concept. Labels were determined by a prosthodontic specialist using the pink esthetic score (PES). A DL model was developed to predict PES based on input images and clinical data. The performance was assessed and compared with that of two other models.

Results: The DL model achieved an average mean absolute error (MAE) of 1.3597, average root mean squared error (MSE) of 1.8324, a Pearson correlation of 0.6326, and accuracies of 65.93 % and 85.84 % for differences between predicted and ground truth values no larger than 1 and 2, respectively. An ablation study demonstrated that incorporating all input features yielded the best performance, with the proposed model outperforming comparison models.

Conclusions: DL demonstrates potential for providing acceptable preoperative PES predictions for single implant-supported prostheses in the esthetic zone. Ongoing efforts to collect additional samples and clinical features aim to further enhance the model's performance.

Clinical significance: The DL model supports dentists in predicting esthetic outcomes and making informed treatment decisions before implant placement. It offers a valuable reference for inexperienced and general dentists to identify esthetic risk factors, thereby improving implant treatment outcomes.

1. Introduction

Implant-supported restoration is a widely used method for addressing dentition defects. In the esthetic zone, particularly the anterior maxilla, achieving both functional and esthetic success is critical for overall treatment outcomes [1]. However, esthetic complications such as soft and hard tissue recession, alveolar ridge resorption, and implant exposure remain common postoperative challenges in this region [2,3]. These complications stem from a variety of factors—including patient characteristics, clinician expertise, biomaterials, and treatment approaches—making it difficult to reliably predict esthetic results [4]. According to Jung and colleagues [5], appropriate implant positioning and ideal clinical attachment levels of adjacent teeth are essential for satisfactory esthetic outcomes in implant treatment. Preoperative

* Corresponding author at: 639 Zhizaoju Road, Shanghai, China.
** Corresponding author at: 800 Dongchuan Road, Shanghai, China.
E-mail addresses: xiaojunchen@sjtu.edu.cn (X. Chen), imxuchun@sjtu.edu.cn (C. Xu).
[1] Ziang Wu and Yizhou Chen contributed equally to this article.

https://doi.org/10.1016/j.jdent.2025.105601
Received 28 October 2024; Received in revised form 28 January 2025; Accepted 30 January 2025
Available online 31 January 2025

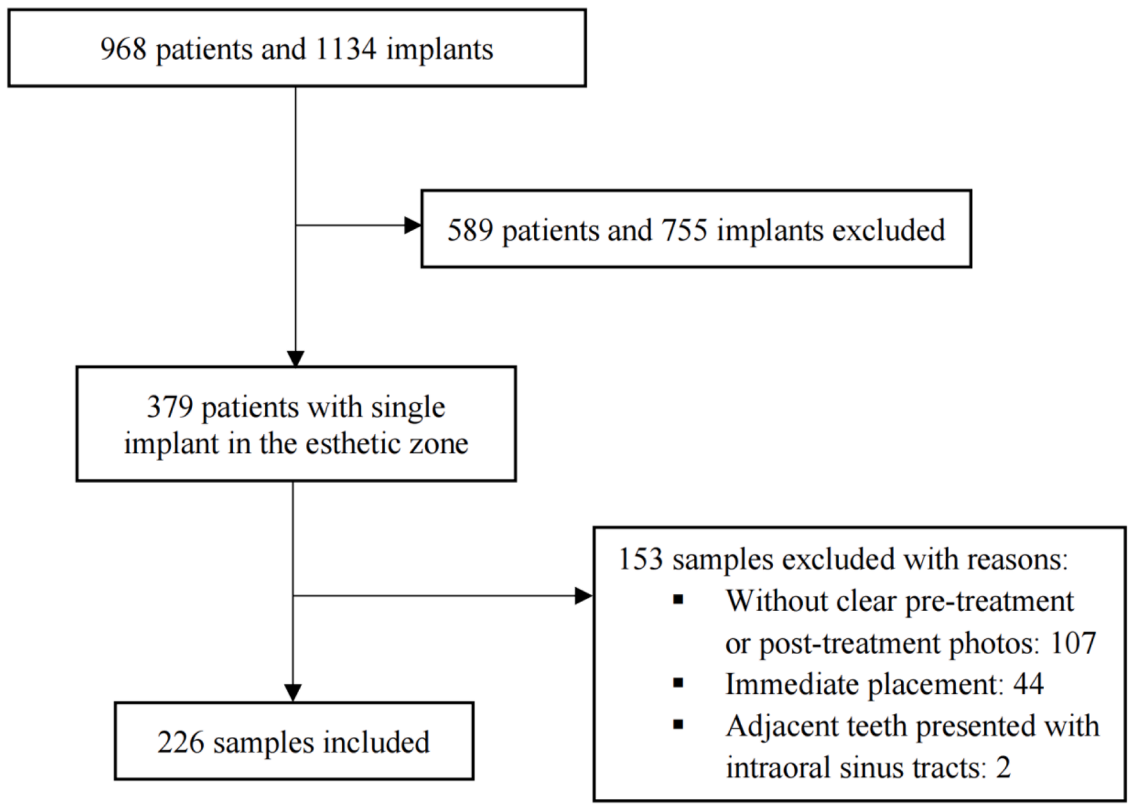

Fig. 1. The flow diagrams for samples.

assessment of peri-implant soft tissue conditions is particularly important for rational decision-making in esthetic cases [6]. The "Straightforward, Advanced, Complex" (SAC) classification system is a practical tool for assessing the risk factors of implant surgery and restoration, categorizing treatment complexity as straightforward, advanced, or complex [7]. Factors such as general health, oral conditions, prosthetic risks, and patient expectations all influence esthetic outcomes to varying degrees. However, the extensive range of risk factors can make this evaluation time-consuming, particularly for general dentists who may lack sufficient implant treatment experience. Consequently, there is a need for supplementary tools to simplify esthetic evaluations.

Recent studies have explored the prediction of esthetic outcomes for complex implant cases. For example, Wang and colleagues [8] examined the relationship between peri-implant phenotypes and esthetic complications, concluding that peri-implant bone and soft tissue components significantly impact implant esthetics and health. They further noted that collecting and assessing all components of the peri-implant phenotype could improve outcome predictability [9]. However, such evaluations require extensive clinical expertise, making them challenging for dental students and less experienced clinicians. Additionally, subjective assessment of treatment factors introduces potential bias, highlighting the necessity of an objective tool for esthetic outcome prediction before implant placement.

Deep learning (DL) has emerged as a powerful tool in dentistry [10], with applications such as caries detection from radiographic images [11,12], anatomical structure segmentation [13], and automatic orthognathic surgery planning [14]. In implant dentistry, DL has been used to identify and classify implants [15,16], segment implants and bone graft materials [17,18], assist with preoperative planning [19,20]. Moreover, DL algorithms have demonstrated promise in predicting implant prognosis, including biological and technical complications, before treatment [21]. Huang and colleagues [22] developed a DL model to predict implant loss using preoperative cone-beam computed tomography (CBCT) and clinical features, achieving strong performance with an area under the receiver operating characteristic curve (AUROC) of 0.90. These studies underscore DL's potential for predicting implant treatment outcomes preoperatively.

Despite these advancements, no DL tools have been proposed for predictive esthetic evaluation of single implant treatments in the esthetic zone. This study aimed to address this gap by developing a DL model for predictive esthetic evaluation of single-implant treatments in this region. The hypothesis was that the model would achieve a mean absolute error (MAE) within 1.5 for the prediction task.

2. Materials and methods

This study adhered to the principles of the Helsinki Declaration (2008 revision) and was approved by the Ethics Committee of Shanghai Ninth People's Hospital (SH9H-2024-T304–1). The study also followed the Checklist for Artificial Intelligence in Dental Research, as detailed in Supplementary Table 1 [23].

2.1. Dataset

To develop the deep learning (DL) model, inclusion and exclusion criteria were defined as follows.

Inclusion criteria:

- Single tooth replacement in the esthetic zone with an implant-supported restoration.
- Two adjacent teeth at the replacement site.
- Clear intra-oral photographs of the maxillary dentition (frontal and occlusal views) before surgery and after final restoration.
- Complete medical and dental history records.
- Early or conventional implant placement following tooth extraction.

Exclusion criteria:

Fig. 2. Intra-oral photographs. (a) Frontal view of the edentulous site. (b) Occlusal view of the edentulous site. (c) Frontal view of the esthetic zone.

- Postoperative intraoral photographs with features (e.g., swelling) that hindered accurate scoring.
- Adjacent teeth with intraoral sinus tracts.
- Immediate implant placement.

From January 2012 to April 2024, 968 implant treatment cases at the Department of Prosthodontics and Second Dental Center, Shanghai Ninth People's Hospital, were screened. Of these, 379 involved single-tooth restorations in the esthetic zone. After applying the inclusion and exclusion criteria, 226 samples were finalized for model training (Fig. 1).

Each sample included three intraoral photographs (Fig. 2) (EOS 5DSR, EF 100/2.8 Macro USM, MT-24EX Twin-flash, Canon, Tokyo, Japan) and eight clinical features related to the patient's medical and dental health. All preoperative photographs were taken 20–30 mins before implant surgery, and postoperative photographs were taken 6 months after the delivery of the final prosthesis. The features included age, gender, smoking status, blood pressure, diabetes, dentition condition (number of missing teeth), adjacent tooth status (e.g., natural tooth, crown, caries, or defect), and implant position. These features were transformed into numeric formats, with mean imputation applied to handle missing data.

Fig. 2. (*continued*).

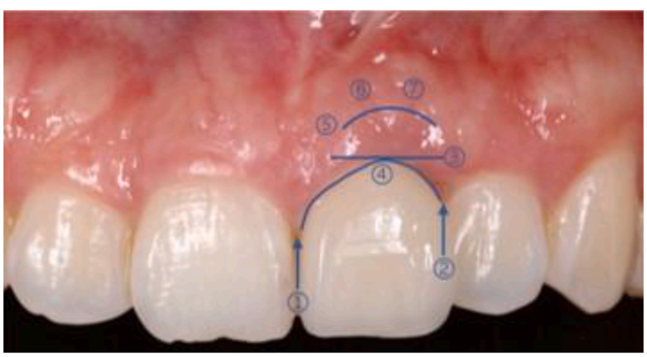

Fig. 3. Brief description of pink esthetic score variables on the intra-oral photograph from the frontal view. 1. Mesial papilla; 2. Distal papilla; 3. Level of soft-tissue margin; 4. Soft-tissue contour; 5. Alveolar process; 6. Soft-tissue color; 7. Soft-tissue texture.

2.2. Labeling and data preprocessing

Esthetic evaluations were based on the pink esthetic score (PES) framework defined by Fürhauser and colleagues [24]. PES assesses seven soft tissue variables: mesial papilla, distal papilla, soft-tissue level, contour, alveolar process deficiency, color, and texture. Each variable is scored on a scale of 0 (worst) to 2 (best), with a maximum total PES of 14.

Two resident dentists independently evaluated a preliminary set of 20 cases (excluded from the main dataset due to missing features) to establish labeling consistency. Inter-rater reliability, assessed using the intraclass correlation coefficient (ICC(A,1)), yielded a high agreement score of 0.954 [25]. Subsequently, the two observers scored the study samples based on PES parameters (Fig. 3). Final labels were reviewed and confirmed by an experienced prosthodontist. The samples were input into the model with labels containing two numbers that represented PES.

Intraoral photographs were annotated using Labelme (CSAIL, MIT, US) before model training. Edentulous areas were marked with rectangular boxes encompassing the missing tooth, adjacent teeth, and gingiva up to the mucogingival junction (Fig. 4). Clinical features were normalized (e.g., age scaled by 100) and converted into binary or integer formats. For example, binary variables represented gender, smoking status, and diabetes, while adjacent teeth status (e.g., tilted teeth, crowns, or gingival recession) was encoded as integers (0, 1, or 2). Implant positions were marked using the FDI numbering system and converted into binary vectors via one-hot encoding.

2.3. Deep learning algorithm

The proposed DL model used a multi-encoder-decoder architecture to predict PES scores from intraoral photographs and clinical features (Fig. 5). It consisted of three identical image encoders, one multi-layer perceptron encoder, three consecutive transformer decoder layers [26], and several fully connected layers. The image encoder was made up of five convolutional layers and five max-pooling layers, ending with a global max-pooling layer, which can encode an image of size $512 \times 512 \times 3$ to a 1024-dimension vector. Three encoders processed the visual information from the intra-oral photographs. The numeric clinical features were concatenated and converted into a 1024-dimension vector after passing through the multi-layer perceptron encoder. These four encoded 1024-dim vectors were stacked and input into the transformer decoder block to capture their mutual correlation. The number of transformer decoder layers was set to 3 and there were 16 heads in the multi-head attention where the hyperparameters were determined after a grid search. The final fully connected layers then decoded and condensed the transformer's output and predicted the PES. The model was trained for 100 epochs with an early stopping strategy using Adam Optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$) with a batch size of 16 and a learning rate of 0.0001. We resized all the input images to 512×512 and applied the RandAugment strategy [27] for data augmentation. Training loss was calculated using mean squared error, with equal weighting for PES variables. The model achieving the lowest validation loss was saved for evaluation.

2.4. Comparison

The predicted results were compared with the ground truth labeled by the observers. Ablation studies were conducted to evaluate the contributions of different input data types, the Transformer block, and data

Fig. 4. Data preprocessing with rectangles using Labelme. (a) Frontal view. (b) Occlusal view.

augmentation strategies. The proposed model was also compared against other baseline models. Given the dataset's limited size, five-fold cross-validation was employed, and the averaged results on test samples were reported.

2.5. Statistical analysis

Model performance was assessed using evaluation metrics for regression tasks, including mean absolute error (MAE), mean squared error (MSE), and accuracy. The evaluation metrics were defined as follows:

$$MAE = \frac{\sum_{i=1}^{N} |\hat{y}_i - y_i|}{N}$$

$$MSE = \frac{\sum_{i=1}^{N} (\hat{y}_i - y_i)^2}{N}$$

$$accuracy = \frac{True}{True + False}.$$

Predicted PES values were initially generated with high decimal precision and rounded to the nearest integer. Accuracy was evaluated based on the proportion of cases where the difference between predicted and ground truth (labeled by dentists) PES values was "True" ≤ 1 (Acc@1). Cases with differences ≥ 2 were considered "False" and incorrect. Additionally, Pearson correlation was calculated to assess the relationship between predicted and ground truth scores.

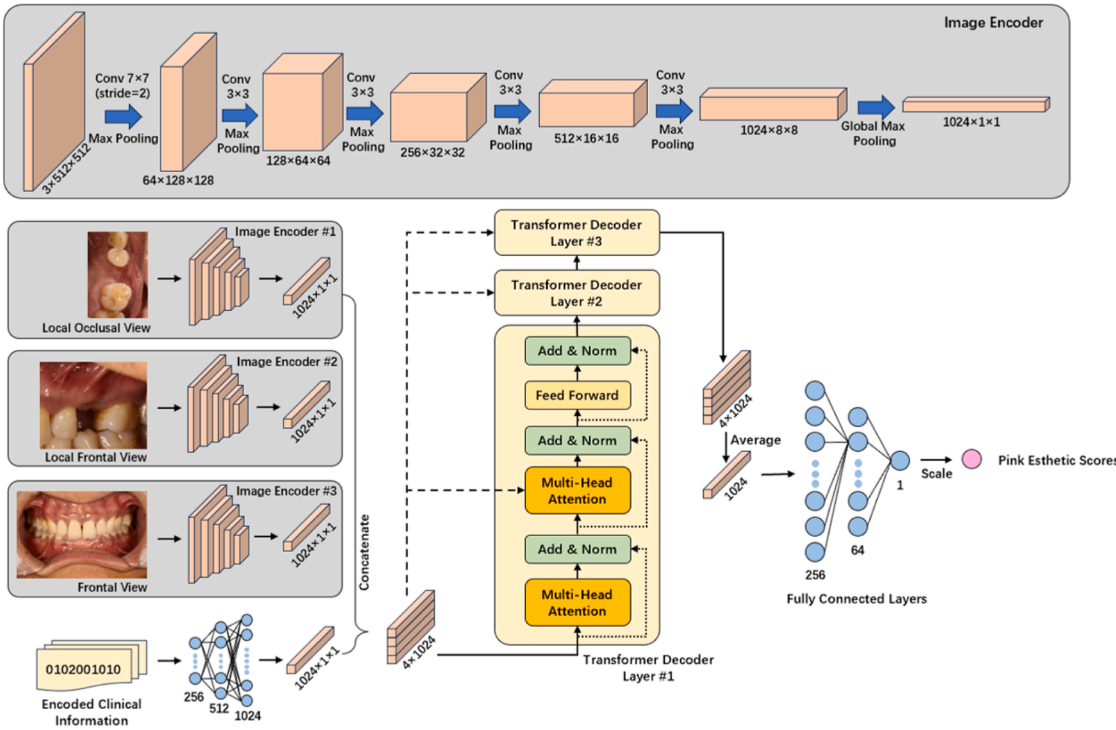

Fig. 5. Overview of the proposed deep learning model architecture for pink esthetic score prediction.

Table 1
Prediction performance of the ablation study.

Input features	Pink esthetic score				
	MAE	MSE	Acc@1	Acc@2	Relevance
w/o local occlusal view	1.4081	1.9089	63.72 %	84.96 %	0.5939
w/o local frontal view	1.3952	1.8857	64.60 %	80.53 %	0.6151
w/o frontal view	1.4016	1.9081	61.50 %	81.86 %	0.5869
w/o clinical features	1.6261	2.0772	57.52 %	80.53 %	0.4331
All included	1.3597	1.8324	65.93 %	85.84 %	0.6326

Abbreviations: w/o, without; MAE, mean absolute error; MSE, mean squared error; Acc@1, accuracy for difference no larger than 1; Acc@2, accuracy for difference no larger than 2.

3. Results

3.1. Performance of the proposed deep learning model

The 226 patients in the study (119 males and 107 females) had ages ranging from 19 to 62 years. The performance of the proposed model was evaluated using five-fold cross-validation, with the averaged results for all test samples summarized in Table 1. The model achieved a mean absolute error (MAE) of 1.3597, mean squared error (MSE) of 1.8324, accuracy@1 of 65.93 %, accuracy@2 of 85.84 %, and a Pearson correlation of 0.6326 for predicting the pink esthetic score (PES) (Fig. 6).

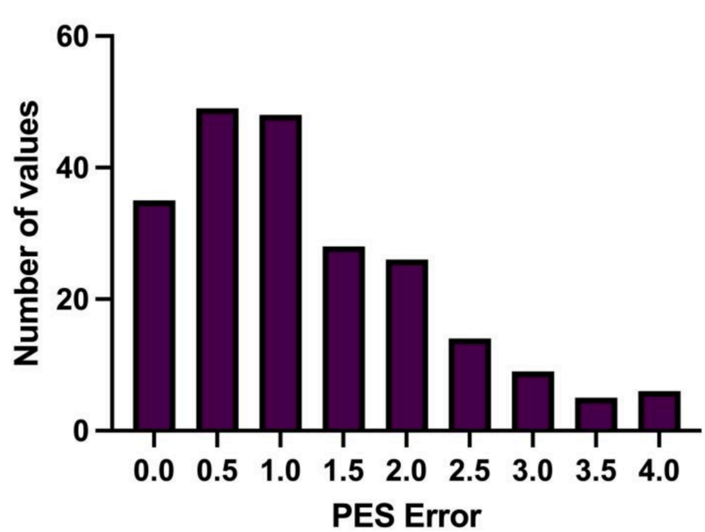

Fig. 6. The frequency histogram of the prediction errors between the proposed model and ground truth.

Table 2
Comparison of prediction performance with other models and settings.

Model/Setting	Pink esthetic score				
	MAE	MSE	Acc@1	Acc@2	Relevance
w/ ResNet18 image encoder	1.4578	1.8793	61.95 %	84.51 %	0.6030
w/ Pretrained image encoder	1.4114	1.8878	62.39 %	83.19 %	0.5819
w/o transformer block	1.4607	1.9184	60.62 %	84.51 %	0.5616
w/o data augmentation	1.4058	1.8677	62.39 %	83.19 %	0.5966
Image size 256×256	1.3829	1.8581	60.18 %	84.96 %	0.5993
Image size 768×768	1.3906	1.8832	63.27 %	83.63 %	0.6041
The proposed model (Image size 512×512)	1.3597	1.8324	65.93 %	85.84 %	0.6326

Abbreviations: w, with; w/o, without; MAE, mean absolute error; MSE, mean squared error; Acc@1, accuracy for difference no larger than 1; Acc@2, accuracy for difference no larger than 2.

3.2. Evaluation based on input features

An ablation study was conducted to assess the impact of different input features on model performance. In each iteration, one branch of the input data was set to null (zero-valued tensor), and the model's performance was measured based on MAE, MSE, and accuracy (Acc@1 and Acc@2, defined by errors ≤ 1 and ≤ 2, respectively). The results are shown in Table 1. The study revealed a significant drop in accuracy when clinical information was excluded from the model, indicating the importance of clinical features for accurate PES prediction. Additionally, incorporating multiple views of oral images further improved prediction accuracy. Notably, the discrepancy between MAE/MSE and Acc@1 arises because accuracy is calculated after rounding predictions to the nearest integer.

3.3. Comparison with other models and settings

Several alternative models and configurations were tested for PES prediction. Table 2 summarizes the comparison results, showing that the proposed model outperformed all alternatives in terms of both MAE/MSE and accuracy. A more sophisticated image encoder, ResNet18 [28], was tested but resulted in an MAE approximately 0.1 higher and an accuracy nearly 4 % lower than the proposed model. Additionally, pretraining the image encoders on a tooth classification dataset (https://huggingface.co/datasets/nekofura/tooth_classification) did not improve PES prediction, leading to higher errors and reduced accuracy. The study also highlighted the essential role of the Transformer block and data augmentation in enhancing model accuracy (Fig. 7). Finally, experiments with different input image sizes revealed that 512 × 512 was the optimal size for this task.

4. Discussion

This study developed a deep learning model to predict the pink esthetic score (PES) after single-tooth implant-supported restoration in the esthetic zone, using pre-operative intra-oral photographs and clinical features. The deep learning algorithm demonstrated clinically acceptable accuracy, with a mean absolute error (MAE) of 1.3597, accuracy@1 of 65.93 %, and a Pearson correlation of 0.6326 for predicting PES. The correlation of 0.6326 indicates a moderate to strong association between the ground truth and the predicted results [29]. The multimodal neural network model outperformed alternative approaches, confirming the hypothesis that the deep learning model would perform with an MAE within 1.5 for PES prediction.

The model was primarily designed to predict PES by evaluating the changes in gingiva before and after implant treatment. The achieved MAE of 1.3597 suggests that deep learning can assess soft tissue outcomes surrounding dental implants. The PES parameters are influenced by both patient [30,31] and implant factors [32]. Shahdad and colleagues [30] found that implant position and facial bone wall thickness impact facial bone structure, influencing PES prediction. Soft-tissue level and alveolar process deficiency (two PES parameters) are closely correlated with preoperative alveolar bone status, which is visible in the intra-oral photographs. Bienz and colleagues [31] also highlighted that the thickness of the soft tissue at implant sites affects esthetic outcomes, particularly the PES parameters related to gingival phenotype. The combination of clinical features and intra-oral photographs provided the model with sufficient data to predict PES with reasonable accuracy.

While PES is commonly used to assess soft tissue outcomes, white esthetic scores (WES), which evaluate implant-supported crowns, were not included in this study. WES, introduced by Belser and colleagues in 2009 [33], focuses on five parameters: tooth form, volume/outline, color, surface texture, and translucency. Unlike PES, WES includes parameters (surface texture and translucency) that are not closely linked to pre-treatment patient factors and cannot be assessed using preoperative photographs. Hence, WES was not part of the deep learning model in this study.

In evaluating the deep learning model, we compared it to ResNet18 and used transfer learning techniques for esthetic prediction. ResNet18, an 18-layer residual network, is designed for classification tasks and has been successfully applied in clinical medicine [34,35]. However, during

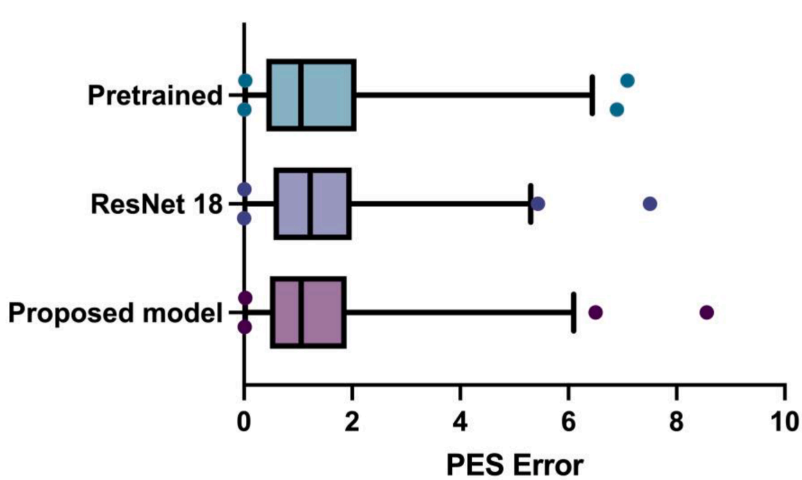

Fig. 7. The box diagram for the comparison of predictive performance between the proposed model, ResNet 18, and pretrained model.

training, the ResNet18 encoders showed signs of overfitting, likely due to the limited dataset size and the larger number of output categories. We also experimented with transfer learning, pretraining the model on a related task, but it did not yield satisfactory results, likely due to the domain mismatch between tasks.

This study is the first to establish a multimodal deep learning model for predicting implant esthetics. The model can predict PES, offering clinicians—particularly those with less experience—an approximate assessment of esthetic outcomes before surgery. Additionally, the predictions align with the labels provided by human evaluators, enabling both clinicians and patients to be informed about potential esthetic results. With only three intra-oral photographs and eight clinical features as input data, the model provides a simple and efficient tool for clinical practice, allowing dentists to save time and make better-informed decisions. More importantly, deep learning serves as an objective tool for assessing the Straightforward, Advanced, Complex (SAC) classification [7] in terms of esthetics. A low output score indicates a more complicated case, suggesting that multidisciplinary treatment may be necessary to improve esthetic outcomes and minimize potential complications. Early access to esthetic predictions also helps patients set realistic expectations.

Despite its advantages, the model has limitations. First, the sample size in the training and testing datasets was relatively small, which may have introduced prediction errors. Zhang and colleagues [36] trained a deep learning model for caries detection using 3932 intra-oral photographs, demonstrating an AUC of 85.65 %, suggesting that a larger sample size could improve performance. Future studies should focus on increasing the number of intra-oral photographs to enhance model accuracy. Second, the intra-oral photographs provide clinicians with 2-dimensional images, and the quality differs between dentists. According to Braun and colleagues, intra-oral scanning, a type of 3-dimensional images, has been applied to esthetic evaluation and showed acceptable reproducibility and reliability compared to the photographs [37]. Future studies might predict the esthetic outcomes based on intra-oral scanning data. Additionally, the clinical features in this study only considered patient factors, without incorporating other critical variables such as treatment strategies, clinician experience, and biomaterials, all of which can impact esthetic outcomes [4,38,39]. Furthermore, radiographic images, especially cone-beam computed tomography (CBCT) scans, could be valuable for PES prediction, as they allow the assessment of alveolar process deficiency, one of the key PES parameters. CBCTs offer more detailed information than 2D intra-oral photographs and could enhance the performance of multimodal deep learning models [40]. Three-dimensional data input could assist the development of multi-modal convolutional neural networks. Finally, all data in this study were collected from Shanghai Ninth People's Hospital, which introduces potential confounding factors due to differences in dentists, cameras, and biomaterials. Future multicenter studies with external validation would enhance the robustness and generalizability of the model [41], providing more reliable predictions for clinical decision-making.

This study focused on conventional implant surgery, but more complex clinical scenarios, such as immediate implant placement (common for the restoration of tooth loss in the esthetic zones) and treatment after consecutive extractions, remain underexplored [42]. Immediate implant placement in the esthetic zone can yield acceptable outcomes, as shown by Chen and Buser [43]. Future deep learning models could predict esthetic outcomes for such cases as well [44]. Additionally, artificial intelligence could help establish standardized evaluation criteria for multiple implants in the esthetic zone, an area that requires further attention and development [45].

5. Conclusions

Despite the limitations of this study, a multimodal neural network model was successfully developed to predict the pink esthetic scores (PES) for single implant-supported restorations based on pre-treatment intra-oral photographs and clinical features. The deep learning model demonstrated the ability to accurately predict esthetic outcomes, offering a valuable preview of the treatment results before implant placement in the esthetic zones. With larger datasets and additional clinical features, the model's performance could be further enhanced, providing more precise clinical guidance for dental implant treatments in the esthetic zone.

CRediT authorship contribution statement

Ziang Wu: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation. **Yizhou Chen:** Writing – review & editing, Visualization, Software, Investigation, Data curation. **Xinbo Yu:** Writing – review & editing, Investigation, Data curation. **Feng Wang:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Haochen Shi:** Writing – review & editing, Software, Investigation, Data curation. **Fang Qu:** Writing – review & editing, Investigation, Data curation. **Yingyi Shen:** Writing – review & editing, Investigation, Data curation. **Xiaojun Chen:** Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Conceptualization. **Chun Xu:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by National Natural Science Foundation (grant number 82071157); Shanghai Municipal Health Commission (grant number 201940009); and Original Exploration Project of Shanghai Ninth People's Hospital, Shanghai Jiao Tong University School of Medicine (grant number JYYC002).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jdent.2025.105601.

References

[1] A. Alrmali, S. Stuhr, M.H.A. Saleh, J. Latimer, J. Kan, D.P. Tarnow, H.-L. Wang, A decision-making tree for evaluating an esthetically compromised single dental implant, J. Esthet. Restor. Dent. 35 (2023) 1239–1248, https://doi.org/10.1111/jerd.13100.

[2] L. Shapira, B.P. Levin, A. Stabholz, Long-term esthetic complications associated with anterior implant-supported restorations, Compend. Contin. Educ. Dent. 42 (2021) 358–363, quiz 364.

[3] A. Ramanauskaite, R. Sader, Esthetic complications in implant dentistry, Periodontol. 2000 88 (2022) 73–85, https://doi.org/10.1111/prd.12412.

[4] S.T. Chen, D. Buser, A. Sculean, U.C. Belser, Complications and treatment errors in implant positioning in the aesthetic zone: diagnosis and possible solutions, Periodontol. 2000 92 (2023) 220–234, https://doi.org/10.1111/prd.12474.

[5] R.E. Jung, L. Heitz-Mayfield, F. Schwarz, Evidence-based knowledge on the aesthetics and maintenance of peri-implant soft tissues: osteology foundation consensus report part 3-aesthetics of peri-implant soft tissues, Clin. Oral Implants Res. 29 (15) (2018) 14–17, https://doi.org/10.1111/clr.13113. Suppl.

[6] W.V. Giannobile, R.E. Jung, F. Schwarz, Evidence-based knowledge on the aesthetics and maintenance of peri-implant soft tissues: osteology Foundation Consensus Report Part 1-Effects of soft tissue augmentation procedures on the maintenance of peri-implant soft tissue health, Clin. Oral Implants Res. 29 (15) (2018) 7–10, https://doi.org/10.1111/clr.13110. Suppl.

[7] A.S. Dawson, S. Chen, D. Buser, L. Cordaro, W.C. Martin, U.C. Belser The SAC Classification in Implant Dentistry, 1st ed., Quintessence Publishing Co, Ltd, Berlin, 2009. https://api.semanticscholar.org/CorpusID:58104216.

[8] I.-C.I. Wang, S. Barootchi, L. Tavelli, H.-L. Wang, The peri-implant phenotype and implant esthetic complications. Contemporary overview, J. Esthet. Restor. Dent. 33 (2021) 212–223, https://doi.org/10.1111/jerd.12709.
[9] G. Avila-Ortiz, O. Gonzalez-Martin, E. Couso-Queiruga, H.-L. Wang, The peri-implant phenotype, J. Periodontol. 91 (2020) 283–288, https://doi.org/10.1002/JPER.19-0566.
[10] X. Cai, H. Zhang, Y. Wang, J. Zhang, T. Li, Digital pathology-based artificial intelligence models for differential diagnosis and prognosis of sporadic odontogenic keratocysts, Int. J. Oral Sci. 16 (2024) 16, https://doi.org/10.1038/s41368-024-00287-y.
[11] E.T. Chaves, S. Vinayahalingam, N. van Nistelrooij, T. Xi, V.H.D. Romero, T. Flügge, H. Saker, A. Kim, G. da S. Lima, B. Loomans, M.-C. Huysmans, F. M. Mendes, M.S. Cenci, Detection of caries around resin-composite restorations on bitewings using deep learning, J. Dent. 143 (2024) 104886, https://doi.org/10.1016/j.jdent.2024.104886.
[12] J. Ver Berne, S.B. Saadi, C. Politis, R. Jacobs, A deep learning approach for radiological detection and classification of radicular cysts and periapical granulomas, J. Dent. 135 (2023) 104581, https://doi.org/10.1016/j.jdent.2023.104581.
[13] H. Wang, J. Minnema, K.J. Batenburg, T. Forouzanfar, F.J. Hu, G. Wu, Multiclass CBCT Image Segmentation for Orthodontics with Deep Learning, J. Dent. Res. 100 (2021) 943–949, https://doi.org/10.1177/00220345211005338.
[14] M. Cheng, X. Zhang, J. Wang, Y. Yang, M. Li, H. Zhao, J. Huang, C. Zhang, D. Qian, H. Yu, Prediction of orthognathic surgery plan from 3D cephalometric analysis via deep learning, BMC. Oral Health 23 (2023) 161, https://doi.org/10.1186/s12903-023-02844-z.
[15] A. Chaurasia, A. Namachivayam, R.B. Koca-Ünsal, J.-H. Lee, Deep-learning performance in identifying and classifying dental implant systems from dental imaging: a systematic review and meta-analysis, J. Periodontal. ImPlant Sci. 54 (2024) 3–12, https://doi.org/10.5051/jpis.2300160008.
[16] M. Revilla-León, M. Gómez-Polo, S. Vyas, B.A. Barmak, G.O. Galluci, W. Att, V. R. Krishnamurthy, Artificial intelligence applications in implant dentistry: a systematic review, J. Prosthet. Dent. 129 (2023) 293–300, https://doi.org/10.1016/j.prosdent.2021.05.008.
[17] B. Tao, J. Xu, J. Gao, S. He, S. Jiang, F. Wang, X. Chen, Y. Wu, Deep learning-based automatic segmentation of bone graft material after maxillary sinus augmentation, Clin. Oral Implants Res. (2023), https://doi.org/10.1111/clr.14221.
[18] B.M. Elgarba, S. Van Aelst, A. Swaity, N. Morgan, S. Shujaat, R. Jacobs, Deep learning-based segmentation of dental implants on cone-beam computed tomography images: a validation study, J. Dent. 137 (2023) 104639, https://doi.org/10.1016/j.jdent.2023.104639.
[19] F.G. Mangano, O. Admakin, H. Lerner, C. Mangano, Artificial intelligence and augmented reality for guided implant surgery planning: a proof of concept, J. Dent. 133 (2023) 104485, https://doi.org/10.1016/j.jdent.2023.104485.
[20] B.M. Elgarba, R.C. Fontenele, F. Mangano, R. Jacobs, Novel AI-based automated virtual implant placement: artificial versus human intelligence, J. Dent. 147 (2024) 105146, https://doi.org/10.1016/j.jdent.2024.105146.
[21] Z. Wu, X. Yu, F. Wang, C. Xu, Application of artificial intelligence in dental implant prognosis: a scoping review, J. Dent. 144 (2024) 104924, https://doi.org/10.1016/j.jdent.2024.104924.
[22] N. Huang, P. Liu, Y. Yan, L. Xu, Y. Huang, G. Fu, Y. Lan, S. Yang, J. Song, Y. Li, Predicting the risk of dental implant loss using deep learning, J. Clin. Periodontol. 49 (2022) 872–883, https://doi.org/10.1111/jcpe.13689.
[23] F. Schwendicke, T. Singh, J.-H. Lee, R. Gaudin, A. Chaurasia, T. Wiegand, S. Uribe, J. Krois, Artificial intelligence in dental research: checklist for authors, reviewers, readers, J. Dent. 107 (2021) 103610, https://doi.org/10.1016/j.jdent.2021.103610.
[24] R. Fürhauser, D. Florescu, T. Benesch, R. Haas, G. Mailath, G. Watzek, Evaluation of soft tissue around single-tooth implant crowns: the pink esthetic score, Clin. Oral Implants Res. 16 (2005) 639–644, https://doi.org/10.1111/j.1600-0501.2005.01193.x.
[25] T.K. Koo, M.Y. Li, A guideline of selecting and reporting intraclass correlation coefficients for reliability research, J. Chiropr. Med. 15 (2016) 155–163, https://doi.org/10.1016/j.jcm.2016.02.012.
[26] A. Vaswani, N.M. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is All you Need. Neural Information Processing Systems, 2017.
[27] E.D. Cubuk, B. Zoph, J. Shlens, Q.V. Le, Randaugment: practical automated data augmentation with a reduced search space, in: 2020 IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops (CVPRW), 2020, pp. 3008–3017, https://doi.org/10.1109/CVPRW50498.2020.00359.
[28] K. He, X. Zhang, S. Ren, J. Sun, Deep residual learning for image recognition, in: 2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778, https://doi.org/10.1109/CVPR.2016.90.
[29] Pearson's correlation coefficient, in: W. Kirch (Ed.), Encyclopedia of Public Health, Springer Netherlands, Dordrecht, 2008, pp. 1090–1091, https://doi.org/10.1007/978-1-4020-5614-7_2569.
[30] S. Shahdad, J. Makdissi, A. Gambôa, Relationship between facial bone dimensions, orofacial implant position, and esthetic outcomes of single-tooth implants, Int. J. Prosthodont. 36 (2023) 668–673, https://doi.org/10.11607/ijp.8052.
[31] S.P. Bienz, M. Pirc, S.N. Papageorgiou, R.E. Jung, D.S. Thoma, The influence of thin as compared to thick peri-implant soft tissues on aesthetic outcomes: a systematic review and meta-analysis, Clin. Oral Implants Res. 33 (23) (2022) 56–71, https://doi.org/10.1111/clr.13789. Suppl.
[32] H. Francisco, D. Marques, C. Pinto, L. Aiquel, J. Caramês, Is the timing of implant placement and loading influencing esthetic outcomes in single-tooth implants?-A systematic review, Clin. Oral Implants Res. 32 (21) (2021) 28–55, https://doi.org/10.1111/clr.13811. Suppl.
[33] U.C. Belser, L. Grütter, F. Vailati, M.M. Bornstein, H.-P. Weber, D. Buser, Outcome evaluation of early placed maxillary anterior single-tooth implants using objective esthetic criteria: a cross-sectional, retrospective study in 45 patients with a 2- to 4-year follow-up using pink and white esthetic scores, J. Periodontol. 80 (2009) 140–151, https://doi.org/10.1902/jop.2009.080435.
[34] Y. Liu, G.-R. She, S.-X. Chen, Magnetic resonance image diagnosis of femoral head necrosis based on ResNet18 network, Comput. Methods Programs Biomed. 208 (2021) 106254, https://doi.org/10.1016/j.cmpb.2021.106254.
[35] Q.A. Al-Haija, M.A. Smadi, S. Zein-Sabatto, Multi-class weather classification using ResNet-18 CNN for autonomous IoT and CPS Applications, in: 2020 International Conference on Computational Science and Computational Intelligence (CSCI), 2020, pp. 1586–1591, https://doi.org/10.1109/CSCI51800.2020.00293.
[36] X. Zhang, Y. Liang, W. Li, C. Liu, D. Gu, W. Sun, L. Miao, Development and evaluation of deep learning for screening dental caries from oral photographs, Oral Dis. 28 (2022) 173–181, https://doi.org/10.1111/odi.13735.
[37] D. Braun, V. Chappuis, M. Fonseca, C. Raabe, V.G.A. Suter, E. Couso-Queiruga, Reproducibility and reliability of intraoral scanners for evaluating peri-implant tissues and implant-supported prostheses: a cross-sectional study, J. Esthet. Restor. Dent. (2025), https://doi.org/10.1111/jerd.13408.
[38] V. Chappuis, M.G. Araújo, D. Buser, Clinical relevance of dimensional bone and soft tissue alterations post-extraction in esthetic sites, Periodontol. 2000 73 (2017) 73–83, https://doi.org/10.1111/prd.12167.
[39] H. Arora, S. Ivanovski, Evaluation of the influence of implant placement timing on the esthetic outcomes of single tooth implant treatment in the anterior maxilla: a retrospective study, J. Esthet. Restor. Dent. 30 (2018) 338–345, https://doi.org/10.1111/jerd.12385.
[40] G. Serindere, E. Bilgili, C. Yesil, N. Ozveren, Evaluation of maxillary sinusitis from panoramic radiographs and cone-beam computed tomographic images using a convolutional neural network, ImAging Sci. Dent. 52 (2022) 187–195, https://doi.org/10.5624/isd.20210263.
[41] Z. Wu, X. Yu, Y. Chen, X. Chen, C. Xu, Deep learning in the diagnosis of maxillary sinus diseases: a systematic review, Dentomaxillofac. Radiol. 53 (2024) 354–362, https://doi.org/10.1093/dmfr/twae031.
[42] D. Buser, V. Chappuis, U.C. Belser, S. Chen, Implant placement post extraction in esthetic single tooth sites: when immediate, when early, when late? Periodontol. 2000 73 (2017) 84–102, https://doi.org/10.1111/prd.12170.
[43] S.T. Chen, D. Buser, Esthetic outcomes following immediate and early implant placement in the anterior maxilla–a systematic review, Int. J. Oral Maxillofac. Implants 29 (2014) 186–215, https://doi.org/10.11607/jomi.2014suppl.g3.3. Suppl.
[44] I. Gamborena, Y. Sasaki, M.B. Blatz, Predictable immediate implant placement and restoration in the esthetic zone, J. Esthet. Restor. Dent. 33 (2021) 158–172, https://doi.org/10.1111/jerd.12716.
[45] J. Cosyn, R. Wessels, R. Garcia Cabeza, J. Ackerman, C. Eeckhout, V. Christiaens, Soft tissue metric parameters, methods and aesthetic indices in implant dentistry: a critical review, Clin. Oral Implants Res. 32 (21) (2021) 93–107, https://doi.org/10.1111/clr.13756. Suppl.